Homework: Comprehensive VHDL Design & Analysis

Instructions: All code must be written in valid VHDL. For drawing tasks, clearly label all signals and components. Follow the "Three Pillars" architectural pattern for all FSM implementations.

Part 1: Conceptual Multiple Choice

Circle the best answer for each question.

1. **In VHDL synthesis, what is the primary purpose of the "Entity" block?**
   a) To describe the internal logical behavior of a circuit.
   b) To define the external interface (ports) of the hardware module.
   c) To specify the physical timing delays of logic gates.
   d) To handle library imports and type definitions.
2. **Which of the following statements about "Blocking Assignments" (:=) is TRUE?**
   a) They should be used for modeling synchronous sequential logic (flip-flops).
   b) They schedule updates to happen at the end of a process execution.
   c) They update variables immediately and are used for intermediate combinational logic.
   d) They are only valid outside of VHDL process blocks.
3. **A Mealy FSM differs from a Moore FSM because its output depends on:**
   a) Only the current state.
   b) Only the current inputs.
   c) Both the current state and the current inputs.
   d) The next state logic and the clock edge.
4. **Why is the process(all) keyword preferred in Modern VHDL for combinational logic?**
   a) It makes the process run faster in actual hardware.
   b) It prevents unintended latches by covering every possible input signal.
   c) It automatically generates the sensitivity list, preventing simulation-hardware mismatches.
   d) It allows the use of non-blocking assignments for variables.
5. **In structural modeling, the port map statement is used to:**
   a) Convert an integer to a STD_LOGIC_VECTOR.
   b) Wire the ports of a component instance to signals in the architecture.
   c) Define the bit-width of a parameterized generic.
   d) Concatenate multiple signals into a single bus.
6. **VHDL requires a specific declaration before you can use a sub-module inside an architecture. This "blueprint" is called a:**
   a) Signal map.
   b) Package body.
   c) Component declaration.
   d) Configuration bit.

Part 2: Drawing and Analysis

Problem 1: Bus Slicing and Structural Logic

Given the following VHDL snippet:

```
lsbmux: mux2 port map(d0(1 downto 0), d1(1 downto 0), s, y(1 downto 0));
msbmux: mux2 port map(d0(3 downto 2), d1(3 downto 2), s, y(3 downto 2));
```

Task: Draw the schematic for this circuit. Show a 4-bit bus entering the module and how it is "sliced" into two 2-bit **mux2** components to produce a 4-bit output **y**.

Part 3: VHDL Coding Tasks

Problem 1: Advanced Combinational Logic (Type Conversion)

Write a VHDL entity and architecture for a **1-to-8 Demultiplexer**.

- **Input:** `din` (single bit), `sel` (`STD_LOGIC_VECTOR(2 downto 0)`).
- **Output:** `dout` (`STD_LOGIC_VECTOR(7 downto 0)`).
- **Requirement:** Use the `TO_INTEGER` function from `IEEE.NUMERIC_STD_UNSIGNED` to route `din` to the index specified by `sel`. All other bits in `dout` must be '0'.

Part 4: Structural Hierarchy

Problem 1: Structural Modeling (4:1 Multiplexer)

Design a **4-input multiplexer** (mux4) by connecting three **2-input multiplexers** (mux2) as components. (Reference: HDL Example 4.14).

- **Ports for** `mux4` : `d0, d1, d2, d3` (4-bit inputs), `s` (2-bit select), `y` (4-bit output).
- **Ports for** `mux2` : `d0, d1` (4-bit inputs), `s` (1-bit select), `y` (4-bit output).
- **Task:**
  1. Write the **Component Declaration** for `mux2` .
  2. Define the **Internal Signals** needed to carry the "low" and "high" intermediate stage results.
  3. Write the **Architecture Body** using three instances of `mux2` (e.g., `lowmux` , `highmux` , `finalmux`) wired together using `port map` .

Problem 4: Hierarchical System Design

You are tasked with building a **16-bit enabled register** (`reg16`) using two **8-bit enabled registers** (`reg8`) as components.

1. **Write the Component Declaration** for `reg8` . It has ports: `clk, reset, en` , `d` (8-bit vector), and `q` (8-bit vector).
2. **Define Internal Signals** (if any) needed to wire the two components together to form a single 16-bit path.
3. **Write the Architecture code** using structural modeling (`port map`) to connect the two `reg8` instances so they share the same control signals (`clk, reset, en`) but handle different halves of the 16-bit data bus.

Part 5

1. **Modeling Styles:** Explain the difference between **behavioral** and **structural** modeling in VHDL. When would it be appropriate to use a structural model instead of a behavioral one?

2. **Inferred Latches:** When using a process to describe combinational logic, what piece of hardware is typically synthesized if you forget to specify the output for all possible input conditions? Why is this generally considered bad practice?

3. **Reset Types:** Describe the key difference in behavior between a synchronous reset and an asynchronous reset for a flip-flop. How does the process sensitivity list change between VHDL implementations of each?

4. **Moore vs. Mealy FSM:** What is the fundamental difference between a Moore and a Mealy finite state machine regarding their outputs? How does this difference affect the

VHDL implementation of the output logic?

5. **Blocking vs. Nonblocking Assignments**: Refer to HDL Example 4.29 (syncbad).
   a) Trace the execution of the two blocking assignments within the process and determine the values of n1 and q after the clock edge.
   b) Based on your trace, explain why this code synthesizes to a single wire or flip-flop instead of the intended two-flop synchronizer.

6. **Structural Model Schematic:** A 4-to-1 multiplexer can be built from three 2-to-1 multiplexers. Draw a logic diagram that represents a structural model of this design.

7. VHDL to Schematic and Analysis: The VHDL code in HDL Example 4.26 uses a nested if/elsif/else structure to describe a priority circuit.
   a) Draw the equivalent logic circuit diagram for this behavioral description.
   b) Explain how the sequential evaluation of an if/elsif/else structure inherently creates priority logic.

8. Parameterized Design and Generate Statements: The VHDL code in HDL Example 4.36 describes a parameterized N-input AND gate using a for...generate statement.
   a) Explain the purpose of the for...generate statement in this VHDL code.
   b) Draw the logic diagram that would be synthesized for the andN entity if the generic parameter width is set to 4.